

Abhishek Jain

Ph.D. | ML Engineer

CONTACT INFORMATION

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🎓 [Publications \(Google Scholar\)](#)

SUMMARY

Controls Ph.D. with 8 years of industry experience designing and deploying AI/ML systems in production. Strong foundation in predictive modeling, constrained optimization, and physical system dynamics. At Tesla, deployed a large-scale real-time data platform actively used by the energy organization's business operations, and leading field trials of a multimodal AI assistant currently being piloted by Tesla's energy service teams.

SKILLS

- **Languages:** Python (PyTorch, FastAPI); MATLAB; C++17, C (hobby projects).
- **AI/ML:** Physics-Based Modeling, Predictive/Prognostic Systems, Deep Learning (CNN, Transformers), LLM/RAG Systems.
- **Physical Systems:** Constrained Optimization, Predictive Control (MPC), Numerical Simulation (RK4), State Estimation.
- **Infrastructure:** Kubernetes (ArgoCD), AWS, PostgreSQL, Vector DBs, Data Pipelines.

PROFESSIONAL EXPERIENCE

Sr. Machine Learning / AI Engineer

March 2025 - Present

Tesla Inc., Bottle Rocket, Palo Alto, CA.

- *Multimodal AI Assistant for Manufacturing:* Initiated and engineered a real-time assistant combining on-device vision, speech recognition, and LLM reasoning for Tesla manufacturing and service technicians; currently in field trials with Tesla's energy service teams.
- *Social Media Analytics Platform:* Conceptualized and launched a global Tesla platform for real-time customer sentiment and market insights, feeding data from social and web sources into a RAG-backed analytics pipeline. Eliminated \$120K/year in third-party tool costs; used directly by VP of Sales/Delivery.
- *Agentic SRE & Tesla Assist:* Led an automated SRE root-cause analysis system using a Reason-Act-Observe loop with real-time tool integrations; separately spearheaded migration of the *tesla.com* RAG backend to Grok-3.

Sr. Machine Learning Engineer

March 2022 - March 2025

J.P.Morgan Chase, Chief Data & Analytics Office, Palo Alto, CA.

- *RAG Chatbot for SRE Productivity:* Developed a generative QA chatbot for SRE teams, achieving 21% reduction in incident tickets and 18% faster resolution times; incorporated production guardrails to minimize hallucinations and an FAQ router for low-latency common queries.
- *Custom LLM Pretraining:* Autonomously trained a GPT-2 style LLM from scratch on 4M internal Confluence documents (5B tokens), achieving BERTScore F1 of 0.65 on JPMC-specific QA.

Sr. Data Scientist

Jan 2018 - Feb 2022

Corning Inc., Manufacturing Data Science Solutions, Corning, NY.

- *Predictive Maintenance for Fiber Draws:* Designed and deployed a Bayesian health metric model and Remaining Useful Life estimator for optical fiber manufacturing equipment — modeling physical degradation trajectories from multi-sensor data streams (temperature, pressure, draw speed) to enable prognostic maintenance scheduling. Data pipeline from PI servers into PostgreSQL; health and RUL predictions surfaced through a real-time monitoring dashboard for plant personnel.
- *Gorilla Glass Defect Classifier:* Led transition from rules-based to hybrid CNN (VGG-19) + Random Forest; achieved >98% recall, 60% reduction in manual inspection, and <10ms inference

via custom C# deployment wrapper. Developed CNN saliency maps for model interpretability, building stakeholder confidence in production predictions.

- *Model Predictive Control*: Designed and deployed MATLAB-based MPC controllers for glass melting furnaces and ceramic filter extrusion lines — system ID via step-response experiments, transfer function derivation, PLC integration, benchmarked against PID on tracking error, settling time, and disturbance rejection across multiple production lines.

Engineering Intern

Summers 2016, 2017

Corning Inc., Manufacturing Technology & Engineering, Corning, NY.

- Emulated PLC control blocks in Matlab/Simulink; analysis improved deployment confidence, increasing deployment rate by 10%.

SIDE QUESTS

- **ARES-Icarus**: Personal physics modeling project — 6-DOF rocket vertical landing simulator in C++17, exploring aerodynamics, numerical integration, and cascade control architectures.
- **ARES-Talos**: Two-wheel self-balancing robot. BNO055 IMU + quadrature encoder fusion for state estimation ($\theta, \dot{\theta}, v$); Embedded C on STM32 for deterministic control.

EDUCATION

Ph.D., Electrical & Computer Engineering

2014 - 2017

North Carolina State University, Raleigh, NC

- Thesis: *Online Distributed Control Designs for Oscillation Damping in Wide-Area Power System Networks*.
- Specialization: Constrained Optimization Methods, Predictive Control Theory, Data Science, Power Networks [GPA: 3.7/4].

M.S., Electrical & Computer Engineering

2011 - 2013

Colorado State University, Fort Collins, CO

- Thesis: *Synchronized Real-Time Simulation of Distributed Networked Controls*.

PATENTS & PUBLICATIONS

- “Predicting optical fiber manufacturing performance using Neural Network,” *U.S. Patent Application No. 16/871,537* (Second Author) [[Patent Link](#)].
- “Advancing SRE Observability with Machine Learning,” *Published on JPMorgan Chase’s tech blog, 2024* (First Author) [[Article Link](#)].
- “Distributed Wide-Area Control of Power System Oscillations under Communication and Actuation Constraints,” *2018 Control Engineering Practice* (First Author) [[Article Link](#)].
- “Structurally Constrained l_1 -Sparse Control of Power Systems: Online Design and Resiliency Analysis,” *2018 American Control Conference* (First Author) [[Article Link](#)].
- “A Model Predictive Control Design for Selective Modal Damping in Power Systems,” *2015 American Control Conference* (First Author) [[Article Link](#)].